



Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E CSE

Course Code & Title: CS3501, Compiler Design

Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima,

AP(SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit 2 / LR-Parser
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Problem Based Learning
- **Justification:**
 - Problem Solving is a basic approach needed for developing any application.
 - **Problem based learning** improves the logical thinking.
 - LR parse table construction helps the students to understand the various parsing techniques and it gives a clear understanding about the syntax analyzer phase.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - Faculty gave a brief explanation about the LR parser. Also, the event was intimated one week in prior and a video lecture prepared by the course handling faculty had been forwarded to their mail. (<https://youtu.be/kzjprndaMKM>).
 - The steps involved in construction of LR parse table (SLR) such as LR items construction and transition among the stages were clearly explained in the video lecture. The students had been asked to watch the video sent to them and take notes.
 - The students formed teams of size four. The teams were given a sample Context free grammar and asked to construct parse table in a paper.
 - They were allowed to discuss among the teammates. Finally one team was called to stage to present their solution.
 - Figure 1 shows the snaps taken while two and three teams (Left and right respectively) were discussing the given problem with their teammates.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO12	PSO1
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CO1	1	3	2	2	1	2
	(1 – Low		2 – Moderate	3 – High)		

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO12	PSO1
Justification for correlation	It involves thinking and make the students to understand clearly what is happening behind the screen while compiling using engineering knowledge	The CFG is analyze thoroughly so that they could construct parse table.	The steps involved in construction of parse table makes the students to analyze the various parsing techniques.	The problem based learning activity involves four members in a team which motivates the team work	The knowledge gained on construction of LR parse table helps the students to understand realworld problem	The students can develop a compiler determining the parsing transition.

• **Images / Screenshot of the practice:**



Fig 1: Glimpse of the activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The students felt that the problem based learning on real world problems are interesting than usual theory classes.
- They felt comfortable to ask doubts with their peers.

❖ **Benefit of the practice:**

- Peer-to-peer learning activities help students develop strong wrapo.
- The students may impart the knowledge they have learned.
- · Knowledge of the subject and retention of it aid students in solving problems in the actual world.

❖ ***Challenges faced in implementation:***

- Assessment on understanding is difficult
- It took more time than planned stipulated time.

References:

- ❖ <https://www.hunschool.org/resources/problem-based-learning.>
- ❖ <https://www.maastrichtuniversity.nl/education/why-um/problem-based-learning.>
- ❖ [https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-\(pbl\).](https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-(pbl).)

Signature of Faculty Member

HOD